

OPERATION MARTINS

Simulated Breach Investigation & SOC Defense Architecture

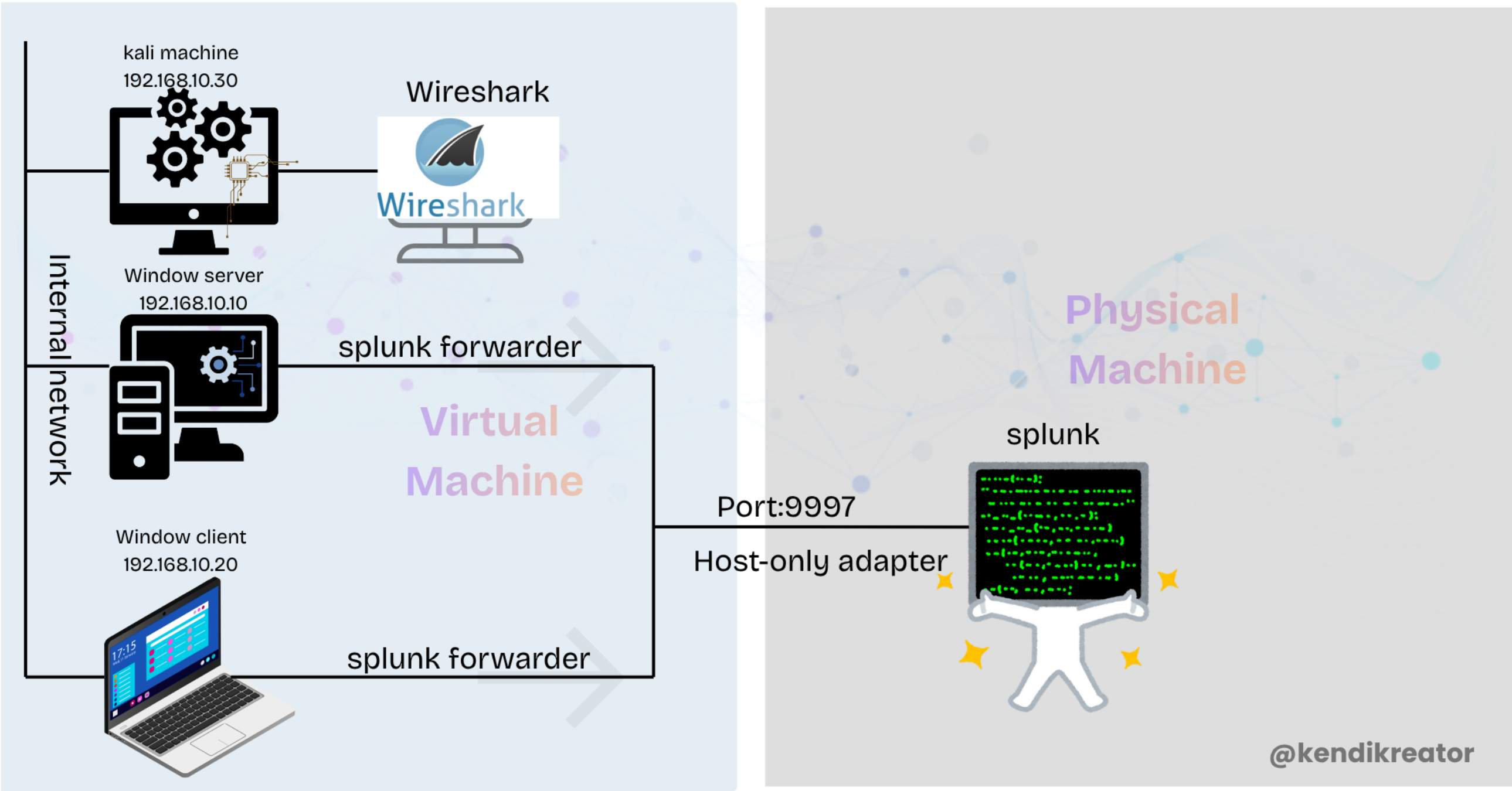
Target Organization: MartinsPay Technologies / Martins Logistics

Environment: Isolated Virtual Testbed (soc-sandbox)

Lead Security Investigator / Defense Presenter

MARTINS JESUROBO PROJECT

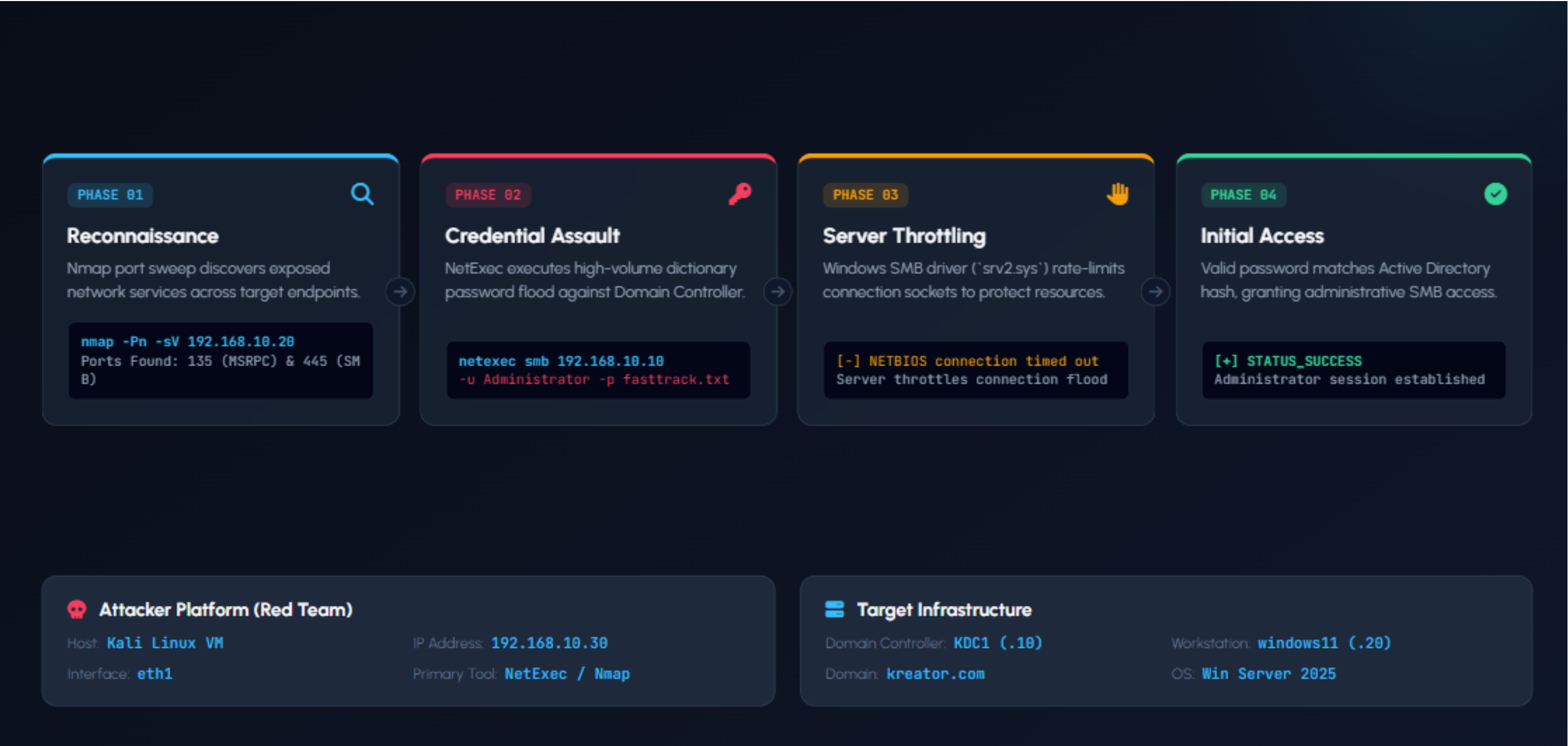
Dual-Adapter Network Architecture & Baseline Topology



Traffic Analysis & Deep Packet Inspection (Wireshark PCAP)

Device / Hostname	IP Address	Expected Protocols & Ports	What Normal Traffic Looks Like
Domain Controller (KDC1)	192.168.10.10	• DNS (53) • Kerberos (88) • LDAP (TCP 389) • SMB (TCP 445)	Workstations requesting domain name resolution, user login tickets, and Group Policy updates over SMB.
Workstation (windows11)	192.168.10.20	• MSRPC (TCP 135) • DHCP / DNS	Standard client endpoint communicating with the Domain Controller to refresh policies and authenticate users.
SIEM Server (Host)	192.168.56.1	• Splunk Ingestion (TCP 9997)	Universal Forwarders on server, kali, workstation streaming Windows Event Logs and linux log to Splunk over the Host-Only adapter.

DELIVERABLE 3 — TASKS 2 & 4: LOG INGESTION & SIEM PIPELINE



DELIVERABLES 5 & 6 — TASKS 6 & 7: THREAT HUNT & INCIDENT TIMELINE

```
(kali@kali)-[~]
└─$ netexec smb 192.168.10.10 -u Administrator -p /usr/share/wordlists/fasttrack.txt
SMB 192.168.10.10 445 KDC1 [*] Windows 11 / Server 2025 Build 26100 x64 (name:KDC1) (domain:kreator.com)
SMB 192.168.10.10 445 KDC1 [-] Connection Error: The NETBIOS connection with the remote host timed out.
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SMB 192.168.10.10 445 KDC1 [-] Connection Error: The NETBIOS connection with the remote host timed out.
```

```
└─$ netexec smb 192.168.10.10 -u Administrator -p [REDACTED]
SMB 192.168.10.10 445 KDC1 [*] Windows 11 / Server 2025 Build 26100 x64 (name:KDC1) (domain:kreator.com) (signing:True) (SMB
SMB 192.168.10.10 445 KDC1 [+] kreator.com\Administrator:[REDACTED] 3d!)
```


Production Detection Analytics & Automated Alert Engineering

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👤 Administrator

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New Search

Save As

Create Table View

Close

index="wineventlog" EventCode=4625

Time range: Today

🔍

🌟

✓ 82,573 events (8/13/26 12:00:00.000 AM to 8/13/26 12:19:52.000 PM)

No Event Sampling

Job

⏸

■

➡

🖨

⬇

💡 Smart Mode

8/13/26

08/13/2026 06:00:14.782 AM

9:00:14.782 AM

LogName=Security

EventCode=4625

EventType=0

ComputerName=KDC1.kreator.com

SourceName=Microsoft Windows security auditing.

Type=Information

RecordNumber=48524

Keywords=Audit Failure

TaskCategory=Logoff

OpCode=Info

Message=An account failed to log on.

Network Information:

Workstation Name: -

Source Network Address: 192.168.10.30

Source Port: 58620

Detailed Authentication Information:

Logon Process: NtLmSsp

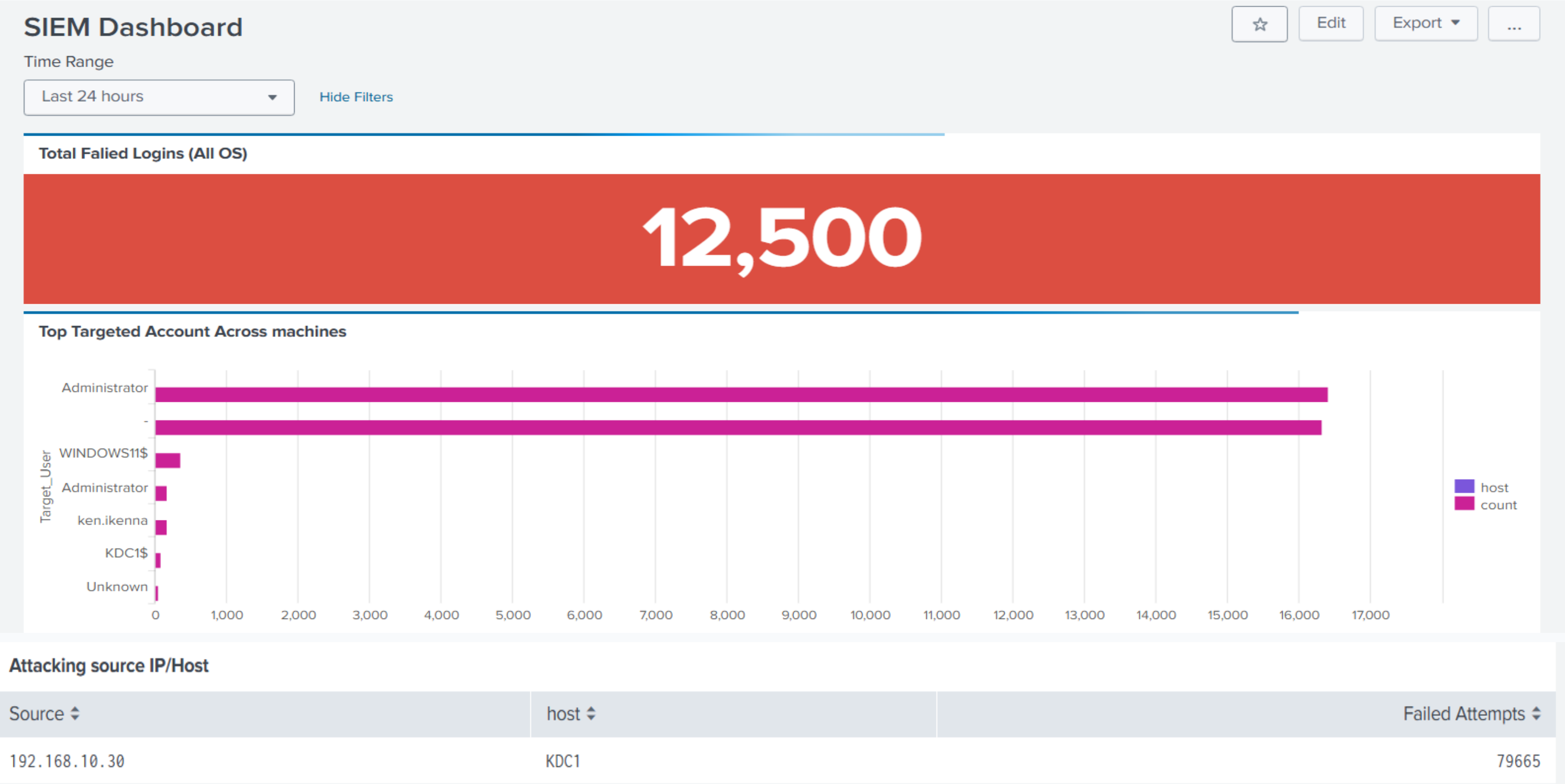
Authentication Package: NTLM

Transited Services: -

Package Name (NTLM only): -

Key Length: 0

Indicators of Compromise (IoCs) & Evidence Matrix



Incident Response Execution

Identification and detection

Containment

Eradication

Recovery

Find it: In **Splunk**, I queried `index=* EventCode=4625` to isolate thousands of failed SMB logon attempts and traced the source IP back to my Kali machine (192.168.10.30).

Stop it: I blocked the Kali IP at the network boundary and disabled the targeted Administrator account in **Active Directory Users and Computers**.

Clean it: I killed active SMB connections, checked Splunk for EventCode 4624 across the network, and verified in Event Viewer that no rogue services (Event 7045) or backdoor users (Event 4720) were installed.

· **Fix it:** I reset the account to a strong passphrase, re-enabled it, and enforced an **Account Lockout Policy** in Group Policy to automatically prevent automated brute-force attacks moving forward."*

